

COLLAFIRTH HILL, United Kingdom

WMO: 030040

Lat: 60.5336N

Lon: 1.3910W

Elev: 748

StdP: 14.30

Time Zone: 0.00 (EUW)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.5	27.3	21.3	16.4	28.5	23.3	18.1	29.6	57.9	35.8	52.0	38.4	19.7	30	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	61.7	56.2	58.0	54.0	55.7	52.9	57.1	60.8	55.2	57.0	53.7	54.9	11.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	67.5	57.7	54.1	64.0	55.7	53.0	61.3	54.3	24.7	60.6	23.6	57.2	22.7	54.8	63.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 51.0	46.0	40.7	DB	22.6	66.7	1.3	3.7	21.6	69.4	20.9	71.5	20.1	73.6	19.2	76.2	(4)
(5)			WB	21.7	60.5	1.2	2.6	20.9	62.3	20.2	63.8	19.5	65.3	18.7	67.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	36.3	35.6	36.1	38.9	44.1	47.6	51.2	51.2	47.3	44.7	39.8	38.2		
	DBStd	6.70	4.31	3.75	3.83	4.05	3.84	3.81	3.10	2.93	3.05	3.20	4.09	4.20		
	HDD50	2833	424	403	430	334	189	92	19	17	88	165	305	367		
	HDD65	8161	889	823	895	784	648	521	428	427	530	629	755	832		
	CDD50	148	0	0	0	0	6	20	56	55	9	2	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	20.0	23.8	22.0	23.2	18.2	17.2	15.3	16.0	16.6	21.1	22.0	21.4	22.7		
	PrecAvg	45.4	5.7	4.4	4.2	2.5	2.3	2.2	2.4	3.2	3.6	4.5	5.1	5.2		
(15) Precipitation	PrecMax	54.9	10.9	6.9	7.3	4.2	4.8	4.3	5.2	6.3	5.5	7.2	7.7	9.2		
	PrecMin	38.6	1.8	2.0	1.5	0.9	0.4	0.4	0.9	1.5	1.1	1.9	2.3	2.0		
	PrecStd	4.6	2.0	1.2	1.1	0.7	0.9	0.9	1.0	1.1	1.1	1.3	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	46.2	45.8	51.4	61.4	65.3	65.5	66.3	56.0	53.4	48.6	47.5		
		MCWB	46.2	44.8	44.9	46.5	51.9	57.4	59.4	58.7	53.6	52.2	48.1	46.8		
	2%	DB	45.4	44.2	44.5	48.0	55.9	60.1	61.1	59.8	54.1	51.2	47.6	46.5		
		MCWB	44.8	43.3	43.8	44.3	48.8	55.0	56.5	55.5	52.5	50.3	47.0	46.0		
	5%	DB	44.0	42.2	43.2	46.1	51.6	55.4	57.6	56.7	52.8	50.2	46.7	45.2		
		MCWB	43.2	41.0	42.2	44.1	46.7	51.5	54.5	54.1	51.6	49.4	46.1	44.6		
	10%	DB	42.3	41.1	42.0	44.5	49.2	52.5	55.3	55.0	51.8	49.1	45.6	44.1		
		MCWB	41.3	39.9	40.8	43.0	45.9	49.7	52.9	53.0	50.8	48.4	44.8	43.3		
	0.4%	WB	46.6	45.2	45.5	47.4	53.7	58.3	59.8	59.8	54.8	52.4	48.5	47.4		
		MCDB	46.6	45.6	45.7	50.3	60.2	63.9	64.0	63.8	54.9	52.7	48.5	47.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.9	43.1	44.0	45.7	49.8	55.0	57.2	56.7	53.1	50.7	47.3	46.2		
		MCDB	45.1	43.7	44.2	46.8	53.5	59.7	60.7	59.5	53.4	51.0	47.4	46.5		
	5%	WB	43.5	41.6	42.6	44.4	47.9	52.1	55.1	55.0	52.1	49.7	46.3	44.9		
		MCDB	43.8	42.2	43.0	45.4	50.7	54.3	56.9	56.1	52.5	50.0	46.6	45.1		
	10%	WB	41.6	40.1	41.1	43.2	46.4	50.6	53.7	53.4	51.1	48.5	45.1	43.6		
		MCDB	42.1	40.9	41.7	44.3	48.5	51.9	54.7	54.3	51.7	48.9	45.6	44.0		
	5% DB	MDBR	4.6	4.9	5.3	6.7	7.4	6.9	6.3	6.0	5.3	5.0	4.5	5.1		
		MCDBR	5.9	6.6	6.2	8.1	11.9	11.5	10.0	8.5	6.2	5.3	4.9	6.6		
		MCWBR	7.4	7.0	6.3	5.6	7.4	6.9	6.6	5.3	5.4	5.3	6.0	7.8		
		MCWBR	6.0	6.6	6.0	7.1	10.5	10.0	8.7	7.7	5.6	5.4	5.0	6.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.5	7.4	6.8	6.2	7.4	6.8	6.8	5.6	5.8	5.7	6.2	7.8		
	Clear-Sky Solar Irradiance	taub	0.317	0.342	0.356	0.369	0.359	0.353	0.366	0.353	0.351	0.335	0.311	0.413		
		taud	2.082	2.224	2.309	2.339	2.428	2.481	2.459	2.495	2.468	2.343	2.134	2.473		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	124	200	242	262	275	279	271	266	245	203	126	83		
		Edh at Noon	15	23	28	33	32	31	31	28	24	20	13	10		
(44) All-Sky Solar Radiation	RadStd		70	226	567	1060	1479	1534	1383	1124	679	339	102	44		
			7	25	40	62	92	166	121	108	62	35	10	3		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)		+0.61	+0.90	+1.30	N/A	N/A	N/A	-167	-221	N/A	N/A

Nomenclature: See separate page